



Assignment 1

DSEZC411 Essential Mathematics for Data Science

Instructions

1. This is not a group assignment. Each student should do the problems and submit individually.
 2. Handwritten assignments have to be scanned and uploaded as a single pdf file with the filename "your BITS-ID".pdf. Add the student BITS-ID in the filename.
 3. Submissions beyond **28th Aug 2026, 11:30 pm** would not be graded.
 4. Assignments sent via email / other electronic forms would not be accepted.
 5. Copying is strictly prohibited. Adoption of unfair means would lead to disciplinary action.
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Answer all the questions

Q1) Consider the set of all vectors $\mathbf{x} \in \mathbb{R}^n$ that satisfies the following condition

$$\sum_{i=1}^n \mathbf{x}_i = 0.$$

Prove that collection of all such vectors is a subspace of $\mathbb{R}^n$.

(1 mark)

Q2) Consider the linear system $\mathbf{Ax} = \mathbf{b}$, where

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 & 4 & 6 \\ 1 & 1 & 7 & 4 & 4 \\ 1 & 5 & 4 & 2 & 1 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} 60 \\ 60 \\ 36 \end{bmatrix}$$

- i) Find the particular solution of $\mathbf{Ax} = \mathbf{b}$ using method learned in lecture.
- ii) Find all the solutions of the associated homegenous linear system.
- iii) Find the expression for the general solution of the linear system.

(2 marks)

Q3) Consider the following matrix

$$\mathbf{C} = \begin{bmatrix} 1 & 2 & 1 & 2 & -1 \\ 1 & 1 & 0 & -1 & 4 \\ 1 & -2 & 4 & 1 & 0 \end{bmatrix}$$

- i) Prove that the space spanned by columns of $\mathbf{C}$ form a subspace of $\mathbb{R}^3$. Find its basis and dimension.
- ii) Prove that the space spanned by rows of $\mathbf{C}$ form a subspace of $\mathbb{R}^5$. Find its basis and dimension.

(2 marks)

Q4) Consider the two subspaces of $\mathbb{R}^3$ named S_1 and S_2 defined as follows:

$$S_1 = \text{span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \right\}$$

$$S_2 = \text{span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$$

- a) Prove that $S_1 \cap S_2$ is a subspace of $\mathbb{R}^3$.
- b) Find the basis and dimension of $S_1 \cap S_2$.

(2 marks)

Q5) Consider the two subspaces of $\mathbb{R}^4$ named S_1 and S_2 defined as follows:

$$S_1 = \text{span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} \right\}$$

$$S_2 = \text{span}\left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Consider the sum operation defined as $S_1 + S_2 = \{\mathbf{x} + \mathbf{y} \text{ where } \mathbf{x} \in S_1 \text{ and } \mathbf{y} \in S_2\}$.

- a) Prove that $S_1 + S_2$ is a subspace.
- b) Find the basis and dimension of $S_1 + S_2$.

(3 marks)